

Acute Effects of Assisted and Resisted Sprint Training on Subsequent Sprint Performance: A Systematic Review and Meta-Analysis

Kai Xu,¹ MingYue Yin,¹ YuMing Zhong,¹ KongYun Huang,² YiMeng Xu,¹ Jing Zhou,² and Ran Wang¹

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Abstract

Xu, K, Yin, M, Zhong, Y, Huang, K, Xu, Y, Zhou, J, and Wang, R. Acute effects of assisted and resisted sprint training on subsequent sprint performance: A systematic review and meta-analysis. *J Strength Cond Res* 39(5): e711–e720, 2025—The aim of the meta-analysis was to determine the acute effects of resisted (RST), assisted (AST), and unresisted (UST) sprint training on sprint performance and to identify the optimal training protocol. A computerized search was conducted in 5 databases, resulting in the inclusion of 23 studies and 395 subjects. The findings indicated that RST acutely improved sprint performance (effect size [ES] -0.20 ; $p < 0.05$), while UST (ES = -0.03) and AST (ES = -0.18) did not produce significant improvements ($p > 0.05$). Subgroup analyses revealed that RST load as a percentage of body mass (%BM) showed the greatest improvement with heavy loads (50–75% BM, ES = -0.40) compared with light (0–19% BM, ES = -0.22), moderate (20–49% BM, ES = -0.21), and very heavy ($>75\%$ BM, ES = 0.10) loads. Further analyses indicated that sled pushing (ES = -0.60) was more effective than sled pulling (ES = -0.34) under heavy load RST conditions. Nonlinear meta-regression results demonstrated that sprint performance improvement exhibited an inverted-U relationship with RST load. In addition, heavy load RST and moderate load AST did not disrupt subsequent sprinting technique. In conclusion, only RST acutely improved subsequent sprint performance, whereas AST and UST did not. For optimal results with RST, it is recommended to use 1 set of heavy loads (50–75% BM) for sled pushing for a distance of 15–20 meters, followed by a rest period of 4–8 minutes before performing 0–30 meters of UST.

Key Words: postactivation performance enhancement, warm-up, complex, kinematics

Introduction

Sprint performance is a critical component in various sports (3,6,63). Faster sprint performance can provide a significant advantage in both team and individual sports (4,23,56). Although numerous training methods have been shown to improve sprint performance (3,63), the principle of specificity suggests that incorporating external resistance or assistance into sprinting may enhance sprint performance more effectively (1,58,85). Consequently, resisted sprint training (RST) and assisted sprint training (AST) are widely used in practice. These methods enhance an athlete's ability to produce greater force and power while maintaining sprint-specific mechanical characteristics, such as movement patterns and contraction types (1,26,58,78). This specificity and transference have the potential to enhance acute sprint performance through a process known as postactivation performance enhancement (PAPE) (17,85).

Postactivation performance enhancement is a strategy that enables acute performance enhancement through submaximal or maximal spontaneous conditioning activities (5,28). Several potential factors, including increased muscle temperature, muscle blood flow, water content, and recruitment of higher order motor

units, may contribute to PAPE (5,60). Traditionally, researchers have used low-volume, high-intensity resistance training (e.g., 1 set of 3 repetitions at 90% 1 repetition maximum [1RM] back squats) to induce PAPE (34). Postactivation performance enhancement can acutely enhance subsequent biomechanically similar power performance (e.g., jumping, sprinting) following a recovery period of 4–12 minutes (66,82). Numerous studies suggested that the greater the similarity between the conditioning activities and the performance tests, the larger the potential for improving PAPE (24,36). Therefore, RST and AST, compared with traditional resistance training, may exhibit greater biomechanical similarities to sprinting and potentially induce a more substantial PAPE (13,26,78). However, the current findings on RST and AST on PAPE are inconsistent. Some studies have found acute enhancement in sprint performance because of RST and AST (31,43,44,54,67–69,81,83,84), while others have found no effect (19,32,35,49,72–74,79) or even impairments (17,19,25,67).

Furthermore, controversy persists regarding the optimal loads for RST and AST. Some researchers (1,39) recommended that the RST load should not exceed 20% of the body mass (BM), and the maximum velocity decrement (V_{dec}) should not exceed 10%, to avoid distorting sprinting technique (23,86). Several meta-analyses on longitudinal RST (1,23,78) suggested that lighter loads ($<20\%$ BM) tend to be more effective than heavier loads ($\geq 20\%$ BM) or generate similar effects. However, PAPE theoretically requires a load of at least 60% 1RM for induction (66,82), suggesting that heavier loads ($\geq 20\%$ BM) are likely to induce PAPE (67,83). Because PAPE is influenced by a balance of

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potentiation and fatigue (5,61), there may be an optimal interval for RST loading: lighter loads may not sufficiently induce PAPE, while heavier loads may cause fatigue. Nevertheless, both lighter and heavier loads (5–150% BM) in RST have shown inconsistent results for PAPE (17,35,49,67,69,72,83). Similarly, Gallego et al. (13) recommended that the maximum velocity in AST should not exceed 110%, but recent studies showed inconsistent results of different AST loads on PAPE (32,35,43,54,68,74). Therefore, determining the optimal loads of RST and AST to induce PAPE remains a challenge.

Considering that complex training typically alternates RST or AST with unresisted sprint training (UST), it can be an effective approach for enhancing sprint performance (1,58,63). Several reviews (1,58,63) and longitudinal intervention studies (16,30,37) have observed the effects of this complex training method on sprint performance improvement. However, the optimal prescription for a single combination (i.e., RST followed by UST or AST followed by UST) for complex training methods remains unclear. Subject characteristics, RST load, RST volume, recovery time, and sprint phase all influence the effects of complex training (66,82). A further breakdown of these factors' impact on sprint performance is necessary to provide coaches and practitioners with more detailed training guidelines. In addition, incorporating RST and AST into the warm-up may produce a greater acute increase in sprint performance based on specific training principles (24,66). Therefore, exploring the acute effects of RST and AST on sprint performance through a comprehensive systematic review is warranted, because a 1% change in athletic performance can determine whether an athlete stands on the podium (60,71). Importantly, there has not been a systematic review and meta-analysis exploring the acute effects of RST and AST on sprint performance.

Given the aforementioned debates and hypotheses, this study aims to determine (1) the acute effect of RST and AST on subsequent sprint performance; (2) influence of subject characteristics (e.g., sex, training level), training program variables (e.g., load, volume), and sprint phase (e.g., 0–10 m, 0–20 m) on pooled results; and (3) the optimal training protocol for RST and AST. In addition, considering that comparisons with UST are commonly made in RST and AST studies, we independently explored the acute effects of UST on sprint performance as an ancillary analysis.

Methods

Experimental Approach to the Problem

This systematic review was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines. The study titles have been registered on the Open Science Framework (osf.io/merts). Electronic literature searches were performed on PubMed, SPORTDiscus, and Web of Science databases up to October 2023. The search strategy used Boolean connectives (OR/AND) with the following expression: (“warm-up” OR “acute” OR “postactivation” OR “short term”) AND ([“sled” OR “partner” OR “elastic” OR “resisted” OR “assisted” OR “vest”] AND [“sprint”]) NOT (“dog”). Additional literature and references were retrieved using Google Scholar and ProQuest. All retrieved studies were compiled and uploaded into EndNote 20 (Clarivate Analytics, Philadelphia, PA), with duplicates later excluded. Two independent reviewers (K. Xu and Y-MY) evaluated the studies based on inclusion and exclusion criteria, with any disagreements were resolved by a third reviewer

(R. Wang) after discussion. The final search results are presented in Figure 1.

Eligibility Criteria

This review considered peer-reviewed English-language studies and dissertations. Inclusion criteria were developed according to the PICOS principle: P (Population) regularly trained healthy individuals, including physically active, amateur athletes, and professional athletes; I (Intervention) RST and AST post-tests with different loads; C (Comparison) RST and AST pretests with different loads; O (Outcome) sprint time and velocity over various distances (e.g., 0–5 m, 0–10 m); S (Study design) randomized controlled, parallel, crossover, and pre-post-test trials. Each study included 1 test result that fits the load classification interval to avoid multiple measurements of the same conditioning activity influencing the meta-analysis (75). If a study reported measurements at multiple time points, the most meaningful outcome was used for calculation (75). Cross-sectional studies, studies without performance pretests, and studies with more than 4 sets of conditioning activities were excluded.

Assessment of Methodological Quality and Risk of Bias

Two reviewers (K. Xu and Y-MY) independently assessed randomized studies using the PEDro scale, evaluating criteria such as random allocation, concealed allocation, baseline between-groups similarity, subjects blinding, therapists blinding, assessors blinding, dropouts, intention-to-treat analysis, between-groups statistical comparison, point measures, and variability data. Eleven criteria were scored 0–1 (0 = no, 1 = yes), excluding criteria 1 and 6 from the review, resulting in a score range of 0–9. The risk of bias for the pre-post-tests was evaluated using the ROBINS-I tool, categorizing bias as “low risk”, “moderate risk,” “serious risk,” “critical risk,” or “no information.”

Data Extraction and Coding of the Study

Data were independently extracted from the included studies by 2 reviewers. Extracted details included the study design, subject information, warm-up design, training equipment, training methods, outcome tests, testing instruments, running surfaces, recovery times, and kinematic changes. Sprint time and velocity results were recorded in an Excel sheet (Microsoft Corporation, Redmond, WA) as mean and SD. For outcome metrics presented as images, data were extracted using WebPlotDigitizer (version 4.5; <https://apps.automeris.io/wpd/>, $r = 0.99$, $p < 0.001$) (21). Corresponding authors were contacted in cases of missing data. Independent variables were categorized as follows: (a) sex: male, female, and mixed groups; athlete category: trained individuals (amateur athletes with regular training and physically active), sprinters and team athletes (e.g., rugby, handball, soccer players); (b) type of conditioning activities: sled pulling and sled pushing; number of conditioning activity sets: single and multiple sets (2–4); RST and AST loads were adapted from Zabalo et al. (86) and categorized as light (0–19% BM), moderate (20–49% BM), heavy (50–75% BM), and very heavy (>75% BM); (c) sprint phase: acceleration phase (0–10 m), transition phase (0–30 m), and maximum velocity phase (>30 m); recovery times: short (0.3–3 minutes), moderate (4–8 minutes), and long (>8 minutes). Coding discrepancies were calculated based on intraclass

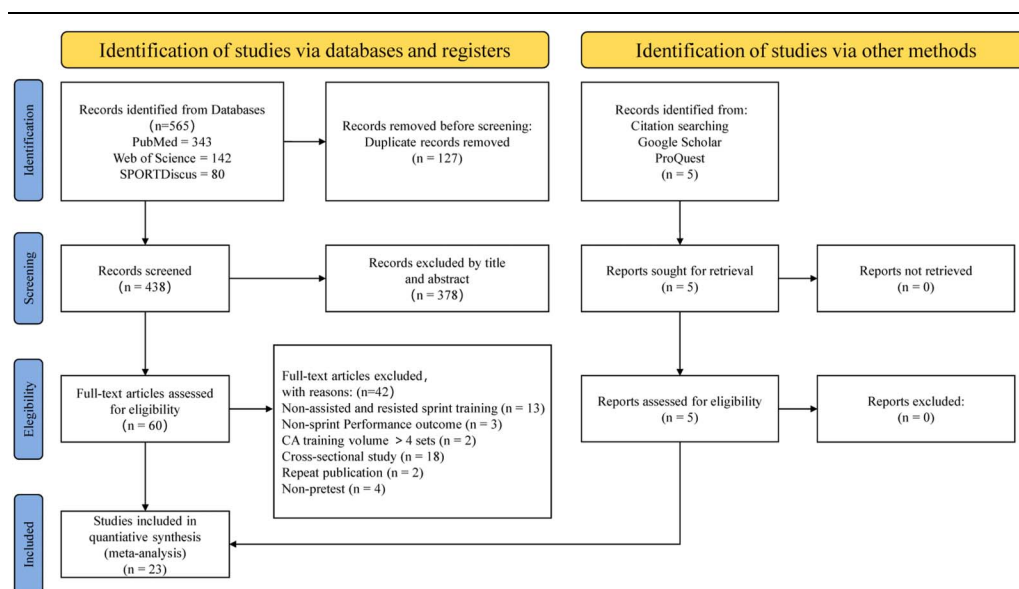


Figure 1. PRISMA flow diagram detailing the study inclusion process.

correlation coefficients (ICC) and an average agreement of 0.90 was considered an appropriate level of reliability. In this study, this value reached 0.92, which ultimately resolved coding discrepancies before the analysis.

Data Synthesis

Analyses were performed using R software (version 4.3.0, R studio). Effect sizes (ES), corrected for small sample sizes using Hedges' and Olkin's g (51), were calculated using the following formula (Equation 1) (29):

$$ES = \frac{(M_{\text{post}} - M_{\text{pre}})}{SD_{\text{pooled}}} \times \left(1 - \frac{3}{4(n_1 + n_2 - 2) - 1}\right), \quad (1)$$

where M_{pre} is the mean of the pretest sprint performance, M_{post} is the mean of the post-test sprint performance, n_1 and n_2 are the sample sizes, and SD_{pooled} is the pooled SD (Equation 2):

$$SD_{\text{pooled}} = \frac{((n_1 - 1) \times SD_1^2 + (n_2 - 1) \times SD_2^2)}{(n_1 + n_2 - 2)}, \quad (2)$$

where SD_1^2 and SD_2^2 are the SD of the performance test completed before and after the conditioning activity, respectively.

For sprint results reported at the average velocity, the conversion was performed by distance/time (1). Effect sizes were categorized using Cohen's scale (18): > -0.2 trivial, $> -0.5 \leq -0.2$ small, $> -0.8 \leq -0.5$ moderate, ≤ -0.8 large. Heterogeneity was tested using I^2 and Q tests, with I^2 categorized as $< 25\%$ (low), $25\text{--}75\%$ (moderate), and $> 75\%$ (high). The Q test was considered significant at $p < 0.1$. Prediction intervals (PI) were calculated to reflect the potential variability of similar studies in the future (7).

Subgroup analyses were conducted based on moderators such as sex and recovery time, exploring the relationship between RST volume (sets $\times$ sprint distance), load, dose (volume $\times$ load %), sprint phase, and PAPE (sprint performance) through linear and nonlinear meta-regression. The RST dose was calculated by taking into account that heavy loads only require less sprint distance, and the multiplication of the 2 results in a more

balanced result. Visualizations were performed using the *ggplot2* package (80) and the *metafor* package (76). Nonlinear meta-regression models were created using Restricted Cubic Spline (27). A random-effects model was used for pooling the main results, and a mixed-effects model was used for subgroup analyses and meta-regression. All analyzed R codes can be found on OSF (osf.io/merts).

Publication bias was assessed by visual inspection of funnel plots and the Egger regression test (22). Sensitivity analyses were conducted to determine the stability of pooled results by eliminating outliers and removing individual ESs one by one (77). The Grading of Recommendations Assessment, Development, and Evaluation (GRADE) methodology was used to assess the level of evidence (65), initially rated as high and downgraded based on sample size (≥ 400), $I^2 > 50\%$, lack of clear merger direction, and risk of publication bias. Grading of Recommendations Assessment, Development, and Evaluation assessments were conducted by 2 authors (K. Xu and Y-MY) and reviewed by a third author (R. Wang). Different populations in the same study were combined by applying Equation (14).

Results

Study Selection and Study Characteristics

The initial database search yielded 565 entries. After removing duplicates, screening titles and abstracts, and evaluating full texts, 18 studies were included. In addition, 5 relevant studies were identified through citations, Google Scholar, and ProQuest, bringing the total to 23 studies (Figure 1). This total comprises 21 RST studies and 6 AST studies. Corresponding data for 2 studies with missing information were provided by the authors (72,87). The 23 included studies used randomized crossover ($n = 16$), parallel ($n = 2$), and pre-post-test trail designs ($n = 5$). In total, these studies involved 395 subjects (male: $n = 292$, female: $n = 103$), consisting of trained individuals ($n = 4$), sprinters ($n = 7$), and team athletes ($n = 12$), aged 16 to 23.2 ± 5.4 years. See Supplemental Digital Content (see Table 1, <http://links.lww.com/JSCR/A595>) for specific details.

Twenty studies conducted warm-up activities above sub-maximal intensity (including general warm-up, dynamic stretching, dynamic exercise, and some potentiation activities), while 3 studies were unclear. Resisted sprint training and AST devices included sleds, 1080 motion, elastic cords, and towing system. Although % V_{dec} is considered more accurate and effective in load selection (1,26,58,78), the load intensities in this study were primarily %BM ($n = 17$), % V_{dec} ($n = 4$), and fixed loads in kilograms ($n = 2$). The load intensities ranged from 5 to 150% of BM. Test distances ranged from 5 to 60 m. Photocells were the primary test instrument ($n = 17$), while some studies used devices such as 1080 motion, velocimeter, laser guns, and cameras. Test venues included indoor ($n = 15$), outdoor ($n = 5$), and unspecified locations ($n = 3$). Recovery times after conditioning activity ranged from 15 seconds to 20 minutes. Kinematic changes in unresisted sprinting after conditioning activity were measured in 7 studies (see Table 2, Supplemental Digital Content, <http://links.lww.com/JSCR/A595>).

Methodological Quality and Risk of Bias

The PEDro scores for randomized trials ranged from 4 to 6, indicating good methodological quality. One study with a high risk of bias was included in the evaluation of pre-post-test studies (68). This study reported only post-test values as percentages without baseline measurements for AST. Despite requesting data from the corresponding authors and receiving no response, we assumed a normal distribution and converted the percentages to sprint times. The other 4 pre-post-test studies had a low risk of bias. Detailed results of the PEDro and ROBINS-I evaluations are provided in the Supplemental Digital Content (see <http://links.lww.com/JSCR/A595>). The funnel plot exhibited symmetry, and the Egger regression test ($p > 0.05$) indicated no significant publication bias (Figure 2).

Meta-Analysis

Meta-analysis revealed a small effect of RST (ES = -0.20 ; 95% confidence intervals [CI] [-0.33 to -0.08]; $p < 0.01$, PI = -0.34 to -0.07 , High GRADE), a trivial effect of AST (ES = -0.18 ; 95% CI [-0.45 to 0.10]; $p = 0.21$, PI = -0.53 to 0.18 , Low GRADE), and UST (ES = -0.03 ; 95% CI [-0.30 to 0.24]; $p = 0.83$, PI = -0.42 to 0.36 , Low GRADE) on subsequent sprint performance (Figure 2). No heterogeneity was observed in any of the 3 pooled results ($I^2 = 0$). Sensitivity analyses showed that the pooled results were robust.

Subgroup Analysis

The load subgroup analysis of the RST (Table 1) indicated that only the H-RST (50–75% BM, ES = -0.40) group had a small significant effect on improving sprinting performance. A single set (ES = -0.22) was more effective than multiple sets (ES = -0.17); moderate recovery time (4–8 minutes, ES = -0.19) was more effective than short (0.3–3 minutes, ES = 0.11) and long (>8 minutes, ES = 0.02) recovery time. The transition phase (0–30 m, ES = -0.19) was more effective than the acceleration phase (0–10 m, ES = -0.13) and the maximum velocity phase (>30 m, ES = -0.16). Female subjects (ES = -0.43) were more effective than the mixed group (ES = -0.22) and male subjects (ES = -0.17), but only the mixed group had a significant effect ($p < 0.05$); team athletes (ES = -0.22) were more effective than trained individuals (ES = -0.20) and sprinters (ES = -0.17). Further analysis of the H-RST subgroups (Table 2) showed that a single set (ES = -0.40) produced a similar significant effect ($p < 0.05$) compared with multiple sets (ES = -0.40). Optimal results were observed for recovery time and test distance in the moderate (4–8 minutes, ES = -0.39) and transition phases (0–30 m, ES = -0.32). Notably, sled pushing (ES = -0.60) was more effective than sled pulling (ES = -0.34).

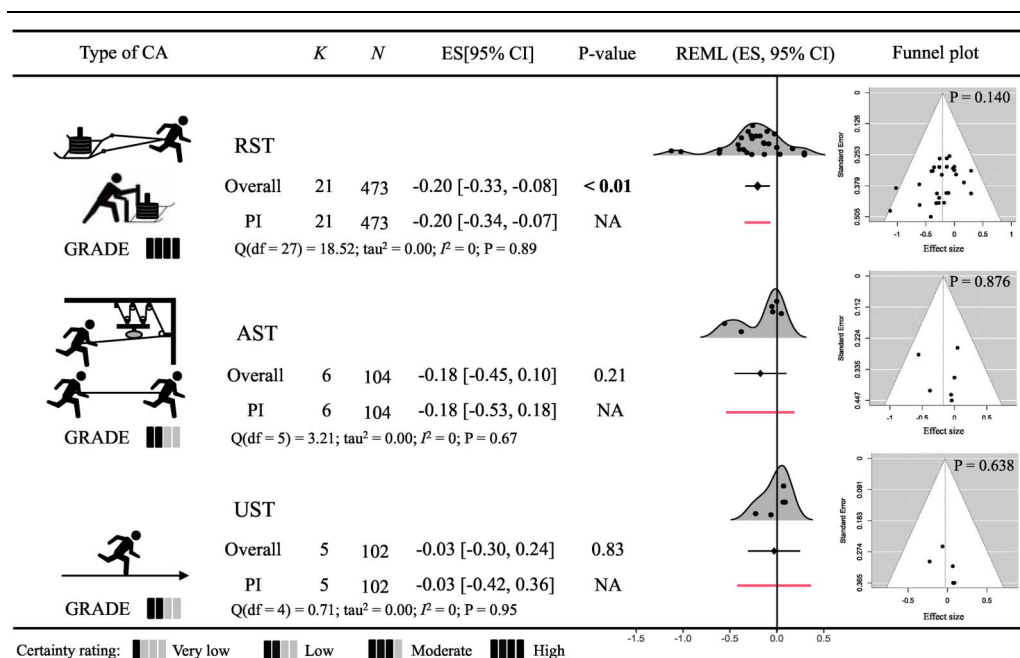


Figure 2. Acute effects of RST, AST, and UST on sprint performance. CA = conditioning activity; K = number of studies; N = sample size; ES = effect size; REML = restricted maximum likelihood; CL = confidence interval; RST = resisted sprint training; AST = assisted sprint training; UST = unresisted sprint training; PI = prediction intervals.

Table 1
Subgroup analysis of different RST and AST.*

Independent variables	Subgroup analysis of RST				Subgroup analysis of AST			
	ES	95% CI	<i>P</i> ²	<i>N</i>	ES	95% CI	<i>P</i> ²	<i>N</i>
Load								
L (0–19% BM)	−0.22	−0.47 to 0.02	0	129	0.01	−0.38 to 0.40	0	51
M (20–49% BM)	−0.21	−0.48 to 0.07	0	102	−0.50	−0.96 to −0.04†	0	38
H (50–75% BM)	−0.40	−0.63 to −0.16†	0	145	0.00	−0.72 to 0.72	0	15
VH (>75% BM)	0.10	−0.18 to 0.38	0	97	NA			
Number of set (s)								
Single	−0.22	−0.38 to −0.05†	0	292	−0.28	−0.67 to 0.10	0	52
Multiple	−0.17	−0.37 to 0.02	0	197	−0.07	−0.45 to 0.32	0	52
Test distance								
0–10 m	−0.13	−0.29 to 0.04	0	296	−0.20	−0.51 to 0.12	0	79
0–30 m	−0.19	−0.34 to −0.04†	0	349	−0.10	−0.44 to 0.25	0	63
>30 m	−0.16	−0.48 to 0.17	0	73	−0.17	−0.65 to 0.32	0	33
Recovery time								
0.3–3 min	0.11	−0.16 to 0.38	0	106	−0.30	−0.68 to 0.09	0	52
4–8 min	−0.19	−0.32 to −0.05†	0	421	0.04	−0.24 to 0.31	0	104
>8 min	0.02	−0.19 to 0.24	0	172	−0.38	−1.19 to 0.43	NA	12
Sex								
Male	−0.17	−0.35 to 0.00	0	261	−0.12	−0.52 to 0.28	0	48
Female	−0.43	−0.88 to 0.01	0	40	−0.25	−0.62 to 0.12	0	56
Mixed	−0.22	−0.43 to −0.01†	0	173	NA			
Type of sports								
Trained	−0.20	−0.49 to 0.10	0	91	−0.15	−1.17 to 0.88	NA	13
Sprinter	−0.17	−0.46 to 0.12	0	93	−0.13	−0.81 to 0.54	0	27
Team	−0.22	−0.38 to −0.05†	0	289	−0.28	−0.83 to 0.27	0	58

*AST = assisted sprint training; BM = body mass; CI = confidence interval; ES = effect size; L, M, H, VH = light, moderate, heavy, very high; NA = not applicable; RST = resisted sprint training.

†Significant difference ($p \leq 0.05$) in sprint performance.

Subgroup analyses of AST (Table 2) indicated that only the M-AST load (20–49% BM, ES = −0.50; $p < 0.05$) had a moderate significant effect on improving sprinting performance. All other moderating variables were not significant ($p > 0.05$).

Linear and Nonlinear Meta-Regression

Given that only 6 studies on AST were available, no meta-regression was performed. To determine the overall effectiveness of RST, no restrictions were imposed on training methods and loads. The present data suggested no significant ($p > 0.05$) linear relationship between RST volume (sets $\times$ sprint distance), load, dose (volume $\times$ load%), sprint phase, and PAPE (Figure 3; see

Figure 1, Supplemental Digital Content, <http://links.lww.com/JSCR/A595>).

The results of the nonlinear meta-regression showed an inverted-U relationship between RST load and PAPE. Resisted sprint training volume and sprint stage at low sprint distances negatively affect sprint performance, gradually improving sprint performance and reaching the upper limit as sprint distance increased. There was no significant relationship between RST dose (volume $\times$ load%) and PAPE (Figure 3; see Figure 1, Supplemental Digital Content, <http://links.lww.com/JSCR/A595>).

Discussion

This meta-analysis aimed to assess the acute effect of RST, AST, and UST on subsequent sprint performance. In addition, it examined moderator such as sex, subject characteristics, number of sets, training volume, recovery time, and sprint phase in the context of RST and AST on PAPE. The ultimate goal was to identify the optimal training protocols for RST and AST. Our main findings suggested (1) RST acutely enhanced subsequent sprint performance, while AST and UST did not; (2) nonlinear meta-regression results indicated that RST showed an inverted-U relationship with load, with less sprint distance in the sprint phase reducing sprint performance and longer distances required to induce PAPE; (3) RST was most effective for the 0–30 m sprint using a single-set heavy load (50–75% BM) push sled with a 4–8 minutes recovery time. Conversely, AST was most effective for the 0–10 m sprint using a single set, moderate load (20–49% BM) with a 0.5–3 minutes recovery time (Figure 4).

This study is the first systematic review and meta-analysis to comprehensively explore the acute effects of RST and AST on sprint performance. A previous meta-analysis (66) synthesizing 47 PAPE studies found that various types of conditioning

Table 2
Subgroup analysis of H-RST.*

	ES	95% CI	<i>P</i> ²	<i>N</i>
Number of set (s)				
Single set	−0.40	−0.70 to −0.09†	0	84
Multiple set	−0.40	−0.72 to −0.07†	0	77
Test distance				
0–10 m	−0.25	−0.54 to 0.04	0	92
0–30 m	−0.32	−0.61 to −0.03†	21%	127
Recovery time				
0–3 min	−0.01	−0.38 to 0.37	0	54
4–8 min	−0.39	−0.63 to −0.16†	0	145
>8 min	−0.13	−0.44 to 0.18	0	80
Type of CA				
Sled pull	−0.34	−0.60 to −0.08†	0	115
Sled push	−0.60	−1.12 to −0.08†	40%	30

*CA = conditioning activity; CI = confidence interval; ES = effect size; RST = resisted sprint training.

†Significant difference ($p \leq 0.05$) in sprint performance.

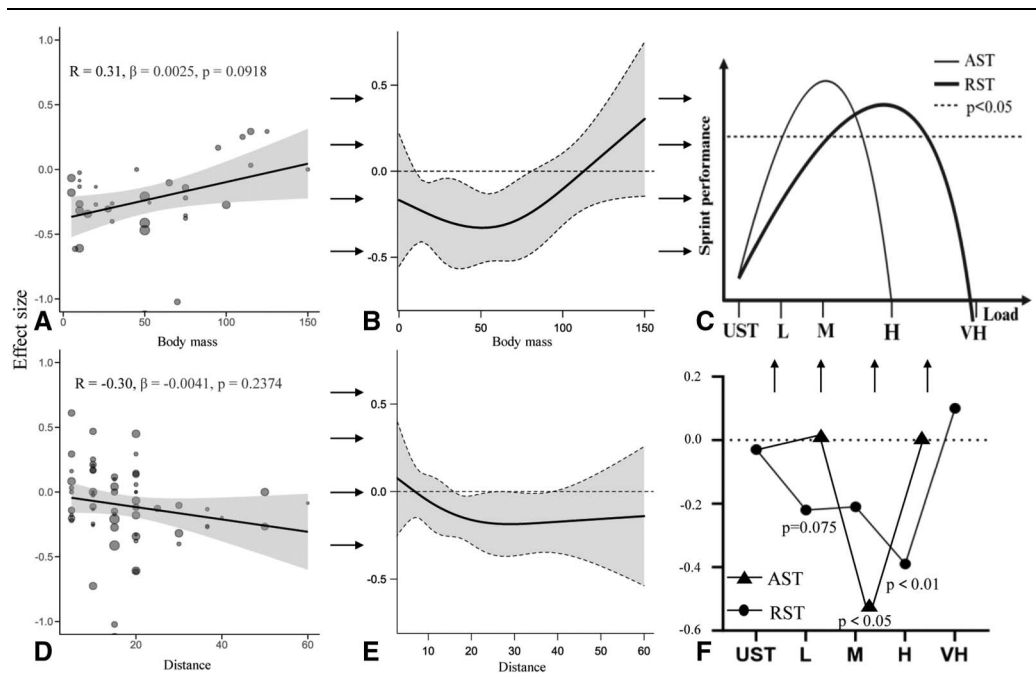


Figure 3. Linear (A) and nonlinear (B) relationship between RST load and PAPE (A), linear (D) and nonlinear (E) relationship between RST sprint distance in the sprint phase and PAPE, pooled effects of UST and different load intensities on sprint performance (F), by extension of the logic of outcome figures B and C, there may be an inverted-U enhancement model of the effect of RST and AST load on PAPE based on the outcome (C). RST = resisted sprint training; AST = assisted sprint training; UST = unresisted sprint training; L, M, H, VH = light, moderate, heavy, very high.

activities, such as resistance training, plyometric, and maximal isometric training produced moderate acute effects on sprint performance. This effect was greater than that observed for RST and AST in this study, suggesting that using specific training methods (i.e., RST and AST) results in a lower-than-average effect compared with nonspecific conditioning activities. However, the CI for ESs in previous meta-analyses were wide (66), and no studies have directly compared the effects of specific training with other conditioning activities. Therefore, it remained clear how specific and nonspecific training differ effects on PAPE.

Furthermore, a recent meta-analysis (40) found no significant effect of different types of conditioning activities (plyometrics, strength-power exercises, RST, and AST) on sprint performance in sprinters ($ES = 0.16, p = 0.08$), indicating that confounders such as types of conditioning activities, subject characteristics, and other factors can affect the magnitude of PAPE.

Among multiple confounders, RST and AST load may play a crucial role in the amplitude of PAPE. Our data showed that only heavy load RST acutely improved subsequent sprint performance. Sprint performance is notably influenced by horizontal

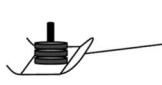
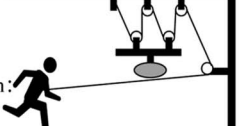
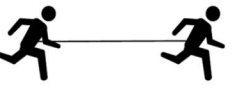
Type of CA	Volume	Recovery	Unresisted sprint
RST			
Sled pull: 	One set /15-20m↑		
Sled push: 	One set /15-20m↑ 50-75%BM↑	4-8min↑	0-30m ↑
AST			
Towing system: 	One set /20-30m↑?		
Elastic cord: 	One set /20-30m↑? 20%-49%BM↑	0.5-3min↑?	0-10m↑?

Figure 4. Overview of the impact of RST and AST on subsequent sprint performance; CA = conditioning activity; RST = resisted sprint training; AST = assisted sprint training; BM = body mass; ↑ = sprint performance improvement; ↑? = largest effect size in subgroup analyses but no significant effect.

forces from ground reaction forces, especially during the acceleration phase (1,23,50,53,78). Kawamori et al. (33) found that heavy loading (30% V_{dec}) compared with light loading (10% V_{dec}) produced more horizontal force. Furthermore, with an increase in RST load, athletes required more contact time and produced more force (8,9), essential for PAPE, which may result in higher neural excitation and muscle temperature, thereby acutely increasing sprint performance (67,83,87). Nevertheless, RST requires a certain load to induce PAPE (66). Neuromuscular stimulation generated by light and moderate loads was insufficient, whereas very heavy load resulted in severe fatigue due to the excess load. Unlike RST, only moderately loaded AST had acute effects on PAPE, possibly because of increased eccentric neuromuscular stimulation during the braking phase of AST compared with UST (13,15,45). Athletes require greater force for bracing, which may activate relevant muscle groups, including the gastrocnemius, quadriceps, biceps femoris, and gluteus maximus (45–47), thereby acutely increasing subsequent sprint performance. However, this result should be interpreted with caution because of the limited number of studies on moderate-load AST.

Another consideration for load selection is that heavier loads in RST ($>20\%$ BM or $10\% V_{dec}$) or AST ($>110\%$ maximum velocity) may disrupt correct sprinting techniques, including kinetics and kinematics. In this study, only 4 studies (17,31,48,87) reported changes in kinematics after RST with more than 20% BM loading. Among them, Jarvis et al. (31) found that a single set of heavy load RST (50% BM) produced a significant change in the hip angle during unresisted sprints ($p < 0.05$, $ES = 0.62$). The authors suggested that the magnitude of hip extension might be attributed to a greater stride length, which was a positive change. However, they found lower consistency for hip extension ($ICC = 0.334$), indicating great variability in athletes' technical movements. The other 3 RST studies (17,48,87) and 1 AST study (74) with loads exceeding 20% BM did not observe a significant effect on subsequent sprint kinematics (step length, step frequency, contact time, and flight time). In addition, several studies (17,41,48,73,79,87) did not observe significant effects on sprint performance kinetics and electromyograms resulting from RST and AST. Therefore, performing single heavier loads of RST and AST did not directly affect subsequent sprinting techniques and patterns.

Based on the results of RST and AST subgroup analyses (Figure 3F) and nonlinear meta-regression of RST loading (Figure 3B), there may be an inverted U-shaped enhancement profile between the RST, AST, and UST (Figure 3C). Numerous cross-sectional studies (2,39,52,57) have shown that sprinting techniques are progressively compromised with increasing RST and AST loads. When the load reaches a certain level and the sprint technique is not entirely compromised, an enhancement (PAPE) window occurs (Figure 3B). The optimal interval for RST in this study was probably between 50 and 75% BM. Assisted sprint training was not subjected to nonlinear meta-regression because of the small number of studies. As the load continued to increase, fatigue increased, biomechanical similarity diminished, and the enhancement of sprint performance gradually decreased until it ultimately vanished. In other words, PAPE requires appropriate loading and biomechanical similarity to the performance tests (83), whereas the same conditioning activity as the test and excessive loading are less likely to induce PAPE (25,67,73,74). Notably, different conditioning activities may produce different inverted U-shaped enhancement profiles for PAPE when applying different loads (e.g., AST and RST, Figure 3B). An extension based on this logic is that all dynamic

conditioning activities have an optimal loading window relative to the individual. For example, back squats induce optimal PAPE only at moderate loads (60–84% 1RM) (66), whereas lighter loads (30% 1RM) require a squat jump to induce PAPE (59). However, it is important to note that these opinions are speculative and need further exploration in future studies.

Another important factor is the relationship between the sprint phase and PAPE. Traditionally, RST (10,11) has been used to enhance the acceleration phase (0–10 m), while AST (13,26) was used to enhance the maximum velocity phase (>30 m). However, our data suggested that RST more acutely improved the transition phase, while AST more acutely improved the acceleration phase. There was significant variation in the overall results of the AST. Simitzi et al. (68) observed that AST provided the greatest enhancement to 0–30 m sprint performance, while Nealer et al. (55) found that AST provided the greatest enhancement to 0–5 m. Considering that only 6 studies were included in the AST, the results of the subgroup analyses in the sprint phase should be interpreted with caution. Many studies (17,19,28,32,48,79,87) have shown that PAPE is difficult to express in the acceleration phase (0–10 m). Owing to the short duration of the early acceleration, it is difficult for potential enhancements to achieve a significant effect. As sprint distance increases, cumulative enhancement is eventually expressed in the 10–30 m interval (17,87). For longer distances, fatigue gradually counteracts the enhancement effect, causing athletic performance to be maintained or decreased. Fernández et al. (20) demonstrated this by comparing the effects of 2 types of conditioning activities with a control group on 0–50 m freestyle swimming performance. Athletes who underwent conditioning activity showed a significant enhancement effect at 0–15 m. However, as distance increased, the conditioning activity group exhibited a noticeable decrease in swimming speed at 35–50 m. Consequently, there was no significant difference in the final 0–50 m performance among the 3 groups. The results of the nonlinear meta-regression on the sprint phase supported these findings, with overall enhancement results showing a weak inverted U shape. Thus, when RST was used as a conditioning activity, PAPE seemed to manifest only in specific sprint phases, with shorter and longer distances likely causing fatigue or having no effect.

Regarding the number of sets, our data showed that both RST and AST were more effective with a single set than in multiple sets. Seitz et al. (66) showed a single set ($ES = 0.44$) was more effective than multiple sets ($ES = 0.21$) in a population of stronger subjects. Athletes were included in 87% ($n = 20$) of the original studies in this review. Although their strength levels were not reported in the studies, previous research has consistently demonstrated the advantage of inducing PAPE in athlete populations (24,28,36,66,70,82). Athlete populations exhibit greater fatigue resistance (62) and higher movement economy (12) than untrained populations when exposed to the same load of conditioning activity because of their consistent and long-term training. Therefore, performing only 1 set of RST and AST was sufficient to induce PAPE.

Regarding recovery time, the optimal recovery time for RST was between 4 and 8 minutes, while AST only required 0.5–3 minutes. Postactivation performance enhancement is influenced by the balance between potentiation and fatigue (64). Compared with UST at the same distance, RST requires greater force generation and more intervention time, potentially leading to greater neuromuscular stimulation (26,67,78,83,87). In contrast, AST requires less energy and intervention time than UST at the same distance (47), potentially leading to less neuromuscular fatigue

(55,74). As a result, RST requires more recovery time, while AST requires less recovery time to improve sprinting performance.

Regarding sex of subjects, our data suggested that both RST and AST had the greatest effect on female subjects. The effect on female subjects in the RST was mainly from Matusiński et al (43,44). They used relatively lighter RST loads (5–15% BM). In contrast, studies (17,19,25,31,41,48,49,67,68,72) with male subjects have mainly used heavier loads (10–150% BM). Considering that male athlete subjects may have greater strength levels than female subjects (28,40,62,66), they may be more susceptible to enhancement effects. Thus, in the RST, female subjects seem to require only relatively light loads, whereas male subjects require relatively heavy loads to induce PAPE. In addition, only 2 studies observed an acute enhancement of AST on sprint performance. One study (68) used 3 sets of 35–40% BM loads on male subjects, and the other (55) used 1 set of 30% BM loads on female subjects. Therefore, a reasonable guess is that male subjects should use multiple sets of moderate loads when using AST, whereas a single set of moderate loads should be used for female subjects. However, considering the small number of studies included in the AST and the lack of significant difference, the true effect of the AST across sex remains unclear.

Regarding the category of athletes, both RST and AST were more effective in team athletes. One possible reason is that sprinters have relatively well-developed sprinting techniques and might be less sensitive to lighter loads. They may require heavier loads and greater neuromuscular stimulation to induce PAPE (72,87). A recent meta-analysis pooling 14 PAPE studies (40) also did not find significant effects of different conditioning activity types in sprinters. It should be noted that in this study, Matusiński et al (43,44), and Simitzi et al. (68) found significant effects of RST and AST in sprinters. However, the former did not report the specifics of the general warm-up, while the latter only used a relatively low-intensity general warm-up. Blazevich and Babault (5) raised concerns about current research on PAPE, suggesting that PAPE is observed only in the absence of adequate warm-up. Therefore, the inconsistent results between different athletes may arise from the warm-up design.

Further analysis of the heavy load RST showed that the number of sets, sprint phase, and recovery time were consistent with the results of overall subgroup analysis of the RST. These results suggest that the overall ES of the RST may be influenced by the heavy load RST. Interestingly, sled pushing had a greater effect on sprint performance than sled pulling. From a kinematic perspective, sled pushing eliminates the arm swing portion, making it less sprint specific than sled pulling. Serrano et al. (42) compared heavily loaded (55 and 90% BM) sled pushing and resisted parachutes. Although sled pushing resulted in more significant kinematic changes than the resisted parachute, it provided greater stimulation to the quadriceps and gastrocnemius muscles, making it more likely to induce PAPE. In conjunction with the theoretical model (Figure 3C), greater neuromuscular stimulation may compensate for the differences in kinematics, producing similar enhancement effects. Therefore, greater neuromuscular stimulation and smaller kinematic changes result in increased PAPE. However, caution is needed when interpreting the results for sled pushing, as only 2 studies have investigated this aspect (25,67).

In interpreting our research findings, several limitations must be mentioned. Five nonrandomized trials (41,68,79,81,84) were systematically included to ensure a comprehensive literature review. One study had a high risk of bias (68), which may have influenced the AST result. Only 6 studies explored AST (32,35,43,55,68,74), so the results of the AST and subgroup

analyses should be interpreted with caution. In addition, only 4 studies (17,48,81,87) used % V_{dec} in the formulation of the loads. Both %BM and fixed loads are influenced by the friction and subject strength levels, without providing an individualized approach that meets each subject's needs (38), which may have implications for the load categorization results. Future studies could delve deeper into loading heavy load RST and moderate load AST to determine whether both are more effective than traditional strength training. Finally, further exploration of the load-biomechanically similar theoretical model (Figure 3B) proposed in this study is warranted to determine the optimal window for each conditioning activity relative to the individual.

Practical Applications

Based on the results of this study, coaches and practitioners should consider integrating RST and AST into complex training regimens and warm-ups to enhance sprint performance. Specifically, RST can be effectively performed with 1 set of heavy loads (50–75% BM) sled pushing for 15–20 m, followed by a 4–8-minute rest before a 0–30 m UST; meanwhile, AST is recommended with 1 set of moderate loads (20–49% BM) partner tows for a 20–30 m sprint, followed by a 0.5–3-minute rest before a 0–10 m UST. Alternating training in this manner may maximize the PAPE effect and optimize long-term sprint performance, and these methods can also be incorporated into a warm-up to acutely enhance subsequent sprints. It should be noted that female athletes might require relatively light loads (~10% BM) when performing RST, and the results further indicated that heavy-load RST and moderate-load AST did not disrupt subsequent sprinting technique—suggesting that coaches need not overly worry about the impact of heavier resisted or assisted sprints on technical movements when selecting RST and AST to induce PAPE.

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References

1. Alcaraz PE, Carlos-Vivas J, Oponjuru BO, Martínez-Rodríguez A. The effectiveness of resisted sled training (RST) for sprint performance: A systematic review and meta-analysis. *Sports Med* 48: 2143–2165, 2018.
2. Bartolini JA, Brown LE, Coburn JW, et al. Optimal elastic cord assistance for sprinting in collegiate women soccer players. *J Strength Cond Res* 25: 1263–1270, 2011.
3. Behrens M, Simonson S. A comparison of the various methods used to enhance sprint speed. *Strength Cond J* 33: 64–71, 2011.
4. Bezodis NE, Willwacher S, Salo AIT. The biomechanics of the track and field sprint start: A narrative review. *Sports Med* 49: 1345–1364, 2019.

5. Blazevich AJ, Babault N. Post-activation potentiation versus post-activation performance enhancement in humans: Historical perspective, underlying mechanisms, and current issues. *Front Physiol* 10: 1359, 2019.
6. Bolger R, Lyons M, Harrison AJ, Kenny IC. Sprinting performance and resistance-based training interventions: A systematic review. *J Strength Cond Res* 29: 1146–1156, 2015.
7. Borg D, Impellizzeri F, Borg S, et al. Meta-analysis prediction intervals are under reported in sport and exercise medicine. *Scand J Med Sci Sports* 34: e14603, 2024.
8. Cahill M, Cronin J, Oliver J, P. Clark K, Lloyd RS, Cross MR. Sled pushing and pulling to enhance speed capability. *Strength Cond J* 41: 94–104, 2019.
9. Cahill M, Cronin J, Oliver J, Clark KP, Lloyd RS, Cross MR. Resisted sled training for young athletes: When to push and pull. *Strength Cond J* 42: 91–99, 2020.
10. Cahill MJ, Oliver JL, Cronin JB, et al. Influence of resisted sled-pull training on the sprint force-velocity profile of male high-school athletes. *J Strength Cond Res* 34: 2751–2759, 2020.
11. Cahill MJ, Oliver JL, Cronin JB, Clark KP, Cross MR, Lloyd RS. Influence of resisted sled-pull training on the sprint force-velocity profile of male high school athletes. *Scand J Med Sci Sports* 30: 442–449, 2020.
12. Cavanagh PR, Kram R. Mechanical and muscular factors affecting the efficiency of human movement. *Med Sci Sports Exerc* 17: 326–331, 1985.
13. Cecilia-Gallego P, Odriozola A, Beltran-Garrido JV, Álvarez-Herms J. Acute effects of overspeed stimuli with towing system on athletic sprint performance: A systematic review with meta-analysis. *J Sports Sci* 40: 704–716, 2022.
14. Chandler J, Cumpston M, Li T, Page MJ, Welch V. *Cochrane Handbook for Systematic Reviews of Interventions*. Hoboken: Wiley, 2019.
15. Clark DA, Sabick MB, Pfeiffer RP, Kuhlman SM, Knigge NA, Shea KG. Influence of towing force magnitude on the kinematics of supramaximal sprinting. *J Strength Cond Res* 23: 1162–1168, 2009.
16. Clark K, Stearne D, Walts C, Miller A. The longitudinal effects of resisted sprint training using weighted sleds vs. weighted vests. *J Strength Cond Res* 24: 3287–3295, 2010.
17. Cochrane DJ, Monaghan D. Using sprint velocity decrement to enhance acute sprint performance. *J Strength Cond Res* 35: 442–448, 2021.
18. Cohen J. A power primer. *Psychol Bull* 112: 155–159, 1992.
19. Crouse CS. *The Acute Effects of Multiple Resisted Sled-Pull Loads on Subsequent Sprint-Running Performances [master's thesis]*. Iowa: University of Northern Iowa, 2015.
20. Cuenca-Fernández F, Ruiz-Teba A, López-Contreras G, Arellano R. Effects of 2 types of activation protocols based on postactivation potentiation on 50-m freestyle performance. *J Strength Cond Res* 34: 3284–3292, 2020.
21. Drevon D, Fursa SR, Malcolm AL. Intercoder reliability and validity of WebPlotDigitizer in extracting graphed data. *Behav Modif* 41: 323–339, 2017.
22. Egger M, Davey Smith G, Schneider M, Minder C. Bias in meta-analysis detected by a simple, graphical test. *BMJ* 315: 629–634, 1997.
23. Fernández-Galván LM, Casado A, García-Ramos A, Haff GG. Effects of vest and sled resisted sprint training on sprint performance in young soccer players: A systematic review and meta-analysis. *J Strength Cond Res* 36: 2023–2034, 2022.
24. Finlay MJ, Bridge CA, Greig M, Page RM. Upper-body post-activation performance enhancement for athletic performance: A systematic review with meta-analysis and Recommendations for future research. *Sports Med* 52: 847–871, 2022.
25. Grimes N, Arede J, Drury B, Thompson S, Fernandes J. The effects of a sled push at different loads on 20 metre sprint time in well-trained soccer players. *Int J Strength Cond* 1, 2021.
26. Hamad MJ, Alcaraz PE, de Villarreal ES. Effects of combined uphill-downhill sprinting versus resisted sprinting methods on sprint performance: A systematic review and meta-analysis. *Sports Med* 54: 185–202, 2024.
27. Harrell FE Jr, Harrell MFE Jr, Hmisc D. *Package "rms" (Vol 229)*. Nashville: Vanderbilt University, 2017.
28. Healy R, Comyns T. The application of postactivation potentiation methods to improve sprint speed. *Strength Cond J* 39: 1–9, 2017.
29. Hedges LV, Olkin I. *Statistical Methods for Meta-Analysis*. New York: Academic Press, 2014.
30. Hicks D, Drummond C, Williams K, van den Tillaar R. The effect of a combined sprint training intervention on sprint force-velocity characteristics in junior Australian football players. *PeerJ* 11: e14873, 2023.
31. Jarvis P, Turner A, Chavda S, Bishop C. The acute effects of heavy sled towing on subsequent sprint acceleration performance. *J Trainol* 6: 18–25, 2017.
32. Jeffreys I. *The Application of Potentiation in Optimizing Warm-Up Procedures in Young Male Athletes [dissertation]*. United Kingdom: University of South Wales, 2010.
33. Kawamori N, Newton RU, Hori N, Nosaka K. Effects of weighted sled towing with heavy versus light load on sprint acceleration ability. *J Strength Cond Res* 28: 2738–2745, 2014.
34. Kilduff LP, Bevan HR, Kingsley MI, et al. Postactivation potentiation in professional rugby players: Optimal recovery. *J Strength Cond Res* 21: 1134–1138, 2007.
35. Kotula K, Matusiński A, Zajac A, Krzysztofik M. Sprint resisted and assisted priming for peak performance. *J Strength Cond Res* 37: 2354–2361, 2023.
36. Krzysztofik M, Wilk M, Stastny P, Golas A. Post-activation performance enhancement in the bench press throw: A systematic review and meta-analysis. *Front Physiol* 11: 598628, 2020.
37. Lahti J, Jimenez-Reyes P, Cross M, et al. Individual sprint force-velocity profile adaptations to in-season assisted and resisted velocity-based training in professional rugby. *Sports* 8: 74, 2020.
38. Lizana JA, Bachero-Mena B, Calvo-Lluch A, et al. Do faster, stronger, and more powerful athletes perform better in resisted sprints? *J Strength Cond Res* 36: 1826–1832, 2022.
39. Lockie RG, Murphy AJ, Spinks CD. Effects of resisted sled towing on sprint kinematics in field-sport athletes. *J Strength Cond Res* 17: 760–767, 2003.
40. Loturco I, Pereira L, Moura T, McGuigan M, Boulosa D. Effects of different conditioning activities on the sprint performance of elite sprinters: A systematic review with meta-analysis. *Int J Sports Physiol Perform* 19: 712–721, 2024.
41. Mangine GT, Huet K, Williamson C, et al. A resisted sprint improves rate of force development during a 20-m sprint in athletes. *J Strength Cond Res* 32: 1531–1537, 2018.
42. Martínez-Serrano A, Marín-Cascales E, Spyrou K, Freitas TT, Alcaraz PE. Electromyography, stiffness and kinematics of resisted sprint training in the specialized SKILLRUN treadmill using different load conditions in rugby players. *Sensors* 21: 7482, 2021.
43. Matusiński A, Golas A, Zajac A, Maszczyk A. Acute effects of resisted and assisted locomotor activation on sprint performance. *Biol Sport* 39: 1049–1054, 2022.
44. Matusiński A, Pietraszewski P, Krzysztofik M, Golaś A. The effects of resisted post-activation sprint performance enhancement in elite female sprinters. *Front Physiol* 12: 651659, 2021.
45. Mero A, Komi PV. Force-, EMG-, and elasticity-velocity relationships at submaximal, maximal and supramaximal running speeds in sprinters. *Eur J Appl Physiol Occup Physiol* 55: 553–561, 1986.
46. Mero A, Komi PV. Electromyographic activity in sprinting at speeds ranging from sub-maximal to supra-maximal. *Med Sci Sports Exerc* 19: 266–274, 1987.
47. Mero A, Komi PV, Rusko H, Hirvonen J. Neuromuscular and anaerobic performance of sprinters at maximal and supramaximal speed. *Int J Sports Med* 8: 55–60, 1987.
48. Monaghan DJ, Cochrane DJ. Can backward sled towing potentiate sprint performance? *J Strength Cond Res* 34: 345–354, 2020.
49. Monahan M, Egan B. The priming effect of a heavy or very heavy resisted sled sprint-based conditioning activity on subsequent unresisted sprint performance in team field sport athletes. *J Strength Cond Res* 37: 91–97, 2023.
50. Morin JB, Bourdin M, Edouard P, Peyrot N, Samozino P, Lacour JR. Mechanical determinants of 100-m sprint running performance. *Eur J Appl Physiol* 112: 3921–3930, 2012.
51. Morris SB. Estimating effect sizes from pretest-posttest-control group designs. *Organ Res Methods* 11: 364–386, 2008.
52. Murray A, Aitchison TC, Ross G, et al. The effect of towing a range of relative resistances on sprint performance. *J Sports Sci* 23: 927–935, 2005.
53. Nagahara R, Mizutani M, Matsuo A, Kanehisa H, Fukunaga T. Association of sprint performance with ground reaction forces during acceleration and maximal speed phases in a single sprint. *J Appl Biomech* 34: 104–110, 2018.
54. Nealer AL, Dunnick DD, Malyszek KK, et al. Influence of rest intervals after assisted sprinting on bodyweight sprint times in female collegiate soccer players. *J Strength Cond Res* 31: 88–94, 2017.
55. Nealer AZ. Influence of rest intervals following assisted sprinting on bodyweight sprint time in female collegiate soccer players. 2016.

56. Nicholson B, Dinsdale A, Jones B, Till K. The training of short distance sprint performance in football code athletes: A systematic review and meta-analysis. *Sports Med* 51: 1179–1207, 2021.
57. Pareja-Blanco F, Pereira LA, Freitas TT, et al. Acute effects of progressive sled loading on resisted sprint performance and kinematics. *J Strength Cond Res* 36: 1524–1531, 2022.
58. Petrakos G, Morin JB, Egan B. Resisted sled sprint training to improve sprint performance: A systematic review. *Sports Med* 46: 381–400, 2016.
59. Poulos N, Haff GG, Nibali M, Graham-Smith P, Newton RU. Comparison of the potentiating effect of variable load jump squats on acute drop jump performance in rugby sevens athletes. *J Strength Cond Res* 37: 149–160, 2023.
60. Prieske O, Behrens M, Chaabene H, Granacher U, Maffiuletti NA. Time to differentiate postactivation “potentiation” from “performance enhancement” in the strength and conditioning community. *Sports Med* 50: 1559–1565, 2020.
61. Rassier DE, Macintosh BR. Coexistence of potentiation and fatigue in skeletal muscle. *Braz J Med Biol Res* 33: 499–508, 2000.
62. Rixon KP, Lamont HS, Bemben MG. Influence of type of muscle contraction, gender, and lifting experience on postactivation potentiation performance. *J Strength Cond Res* 21: 500–505, 2007.
63. Rumpf MC, Lockie RG, Cronin JB, Jalilvand F. Effect of different sprint training methods on sprint performance over various distances: A brief review. *J Strength Cond Res* 30: 1767–1785, 2016.
64. Sale DG. Postactivation potentiation: Role in human performance. *Exerc Sport Sci Rev* 30: 138–143, 2002.
65. Schünemann HJ, Higgins JP, Vist GE, et al. Completing “Summary of findings” tables and grading the certainty of the evidence. *Cochrane Handbook Systematic Reviews Interventions*: 375–402, 2019.
66. Seitz LB, Haff GG. Factors modulating post-activation potentiation of jump, sprint, throw, and upper-body ballistic performances: A systematic review with meta-analysis. *Sports Med* 46: 231–240, 2016.
67. Seitz LB, Mina MA, Haff GG. A sled push stimulus potentiates subsequent 20-m sprint performance. *J Sci Med Sport* 20: 781–785, 2017.
68. Simitzi V, Tsoukas A, Kostikiadis I, et al. The acute effects of different high-intensity conditioning activities on sprint performance differ between sprinters of different strength and power characteristics. *Kinesiology* 53: 193–205, 2021.
69. Smith CE, Hannon JC, McGladrey B, Shultz B, Eisenman P, Lyons B. The effects of a postactivation potentiation warm-up on subsequent sprint performance. *Hum Mov* 15: 36–44, 2018.
70. Suchomel TJ, Lamont HS, Moir GL. Understanding vertical jump potentiation: A deterministic model. *Sports Med* 46: 809–828, 2016.
71. Taylor M, Beneke R. Spring mass characteristics of the fastest men on earth. *Int J Sports Med* 33: 667–670, 2012.
72. Thompson KMA, Whinton AK, Ferth S, Spriet LL, Burr JF. Moderate load resisted sprints do not improve subsequent sprint performance in varsity-level sprinters. *J Strength Cond Res* 35: 72–77, 2021.
73. Van Den Tillaar R, Teixeira A, Marinho D. Acute effect of resisted sprinting upon regular sprint performance. *Acta Kinesiol Univ Tartu* 23: 19–33, 2018.
74. Van Den Tillaar R, Von Heimburg E. Comparison of different sprint training sessions with assisted and resisted running: Effects on performance and kinematics in 20-m sprints. *Hum Mov* 18: 21–29, 2018.
75. Vasconcelos GC, Brietzke C, Cesario JCS, et al. No evidence of post-activation performance enhancement on endurance exercises: A comprehensive systematic review and meta-analysis. *Med Sci Sports Exerc* 56: 315–327, 2024.
76. Viechtbauer W. Conducting meta-analyses in R with the metafor package. *J Stat Software* 36: 1–48, 2010.
77. Viechtbauer W, Cheung MWL. Outlier and influence diagnostics for meta-analysis. *Res Synth Methods* 1: 112–125, 2010.
78. Ward C, Cathain C, Chéilleachair NN, Grassick S, Kelly DT. Does resisted sprint training improve the sprint performance of field-based invasion team sport players? A systematic review and meta-analysis. *Sports Med* 54: 659–672, 2024.
79. Whelan N, O'Regan C, Harrison AJ. Resisted sprints do not acutely enhance sprinting performance. *J Strength Cond Res* 28: 1858–1866, 2014.
80. Wilkinson L. In: *ggplot2: Elegant Graphics for Data Analysis*. WICKHAM H, ed. Oxford: Oxford University Press, 2011.
81. Williams JJ, Herron RL, Spradley B, Saracino P. Postactivation potentiation effect of heavy sled towing on subsequent sprints. *J Strength Cond Res* 35: 1229–1233, 2021.
82. Wilson JM, Duncan NM, Marin PJ, et al. Meta-analysis of post-activation potentiation and power: Effects of conditioning activity, volume, gender, rest periods, and training status. *J Strength Cond Res* 27: 854–859, 2013.
83. Winwood PW, Posthumus LR, Cronin JB, Keogh JWL. The acute potentiating effects of heavy sled pulls on sprint performance. *J Strength Cond Res* 30: 1248–1254, 2016.
84. Wong MA, Dobbs IJ, Watkins CM, et al. Sled towing acutely decreases acceleration sprint time. *J Strength Cond Res* 31: 3046–3051, 2017.
85. Young WB. Transfer of strength and power training to sports performance. *Int J Sports Physiol Perform* 1: 74–83, 2006.
86. Zabaloy S, Freitas TT, Pareja-Blanco F, Alcaraz PEE, Loturco I. Narrative review on the use of sled training to improve sprint performance in team sport athletes. *Strength Cond J* 45: 13–28, 2023.
87. Zizi M, Stavridis I, Agilara GO, Economou T, Paradisis G. The acute effects of heavy sled towing on acceleration performance and sprint mechanical and kinematic characteristics. *Sports* 10: 77, 2022.